UK Greenhouse Gas Emissions: Trend Analysis and 5-Year Forecast

1. Data and Preparation

This analysis is based on the ONS Provisional Atmospheric Greenhouse Gas Emissions dataset, covering annual UK emissions from 1990 to 2024, measured in thousand tonnes of CO_2 equivalent.

The source dataset was not analysis ready. It contained multiple header rows, notes, multiple tables in a single sheet and non-numeric placeholders such as "[low]".

The following steps were carried out:

- Identified the correct data header using the column labelled "Industry name", which contains the years.
- Removed metadata rows and unwanted columns.
- Split each sheet into two tables (industry sections and industry groups) and cleaned both.
- Replaced non-numeric placeholders with zeros and converted numeric fields. (Although the ONS documentation explains these values are not truly zero, converting them allowed the dataset to remain numerically consistent and usable for modelling)
- Extracted a clean annual time series of **total UK emissions** for modelling.
- Saved the cleaned datasets as CSV files for reuse.

This produced a consistent national emissions dataset from 1990 to 2024.

2. Modelling Approach

To forecast the estimated amount of greenhouse gases in the UK over the next five years, I apply both the Standard model and Damped Trend model of **Holt's Linear Trend Exponential Smoothing method**.

- The choice of this method is based on the nature of the data, where the data consists of an annual series and has a long-term trend of declining **without seasonality**.
- This data is suited for the Holt method since the model can predict the level as well as the trend of the series while being stable even with small data.
- The **dampened version** of the model was employed **to prevent the projection of overly aggressive reductions in the future**, especially after the shock exhibited during the COVID-19 era (2020-2021), as seen in Fig 1. The dampening

effect gradually reduces the effect given by the estimated trend, making it realistic.

Assumptions

- The long-term downward trend continues.
- No major structural policy or economic shock occurs.
- **The years 2020 and 2021 are treated with caution due to COVID-19**, as shown by the sharp dip in **Fig 1**, because this unusual period could otherwise mislead the forecast.

3. Findings

Historical Trend (Fig 1)

Fig 1 shows that UK greenhouse gas emissions have **fallen steadily** since 1990.

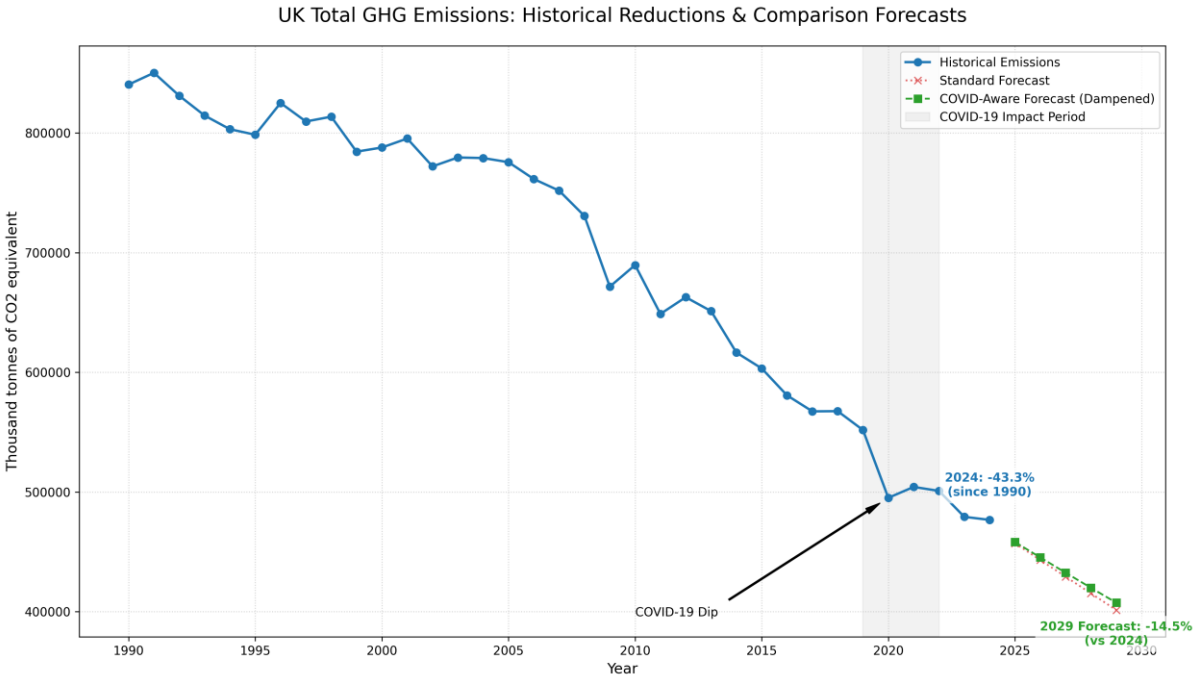


Fig 1: UK Total GHG Emissions: Historical Reductions & Comparison Forecasts

By **2024**, emissions were approximately **43% lower than in 1990**, showing long-term progress.

The chart also highlights the sharp fall in 2020 during COVID-19, followed by partial recovery. This confirms the need for the **dampened model** rather than relying purely on recent data.

5-Year Forecast (Fig 1)

Both models predict a continued decline in emissions between 2025 and 2029. The dampened model projects emissions in **2029 to be about 14–15% lower than in 2024**, indicating that **progress will continue but at a slower pace**.

This suggests that current policies are helping but are **not reducing emissions fast enough on their own** to guarantee reaching net zero by 2050.

Sector Level Insights (Fig 2 & Fig 3)

Fig 2 (donut chart) shows the emissions breakdown for 2024. The largest contributors are:

- Consumer expenditure (26%)
- Transport and storage (16.1%)
- Electricity, gas and steam supply (14.9%)
- Manufacturing (13.7%)
- Agriculture (10.2%)

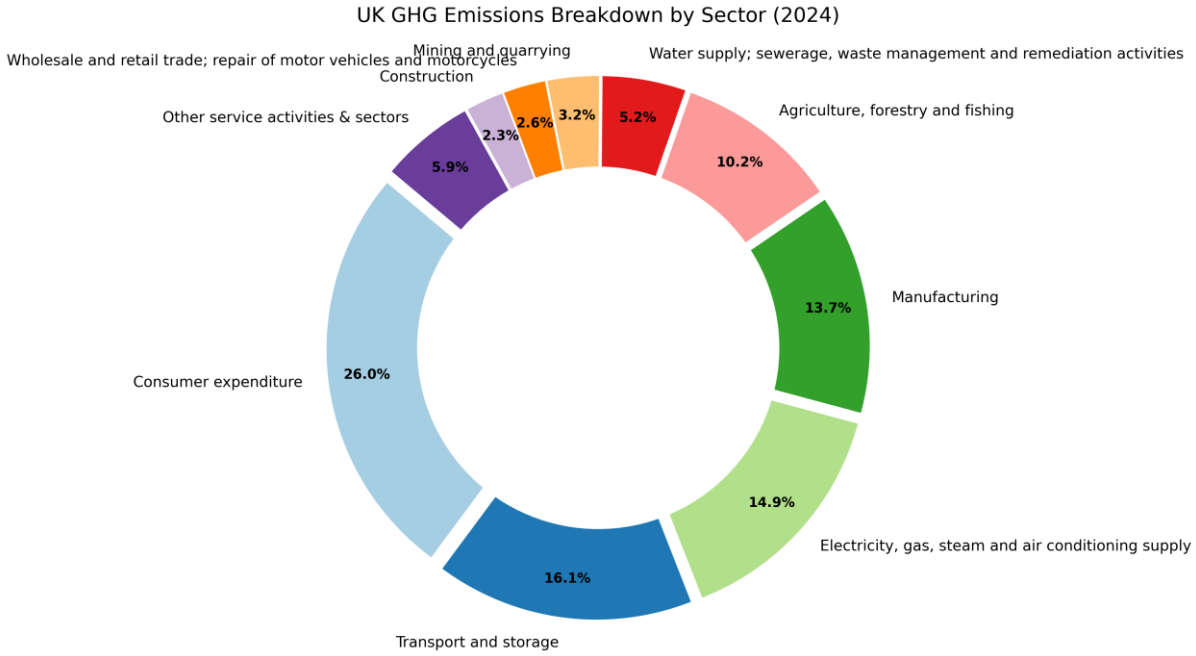


Fig 2: UK GHG Emissions Breakdown by Sector (2024)

Together, these sectors account for a major part of the UK emissions, meaning policy impact will be greatest if focused here.

Fig 3 shows that consumer expenditure (travel and non-travel) dominates, followed by agriculture-related products, air transport services, and electricity production. This

reinforces the message from Fig 2; a small number of sectors and activities drive a large share of emissions.

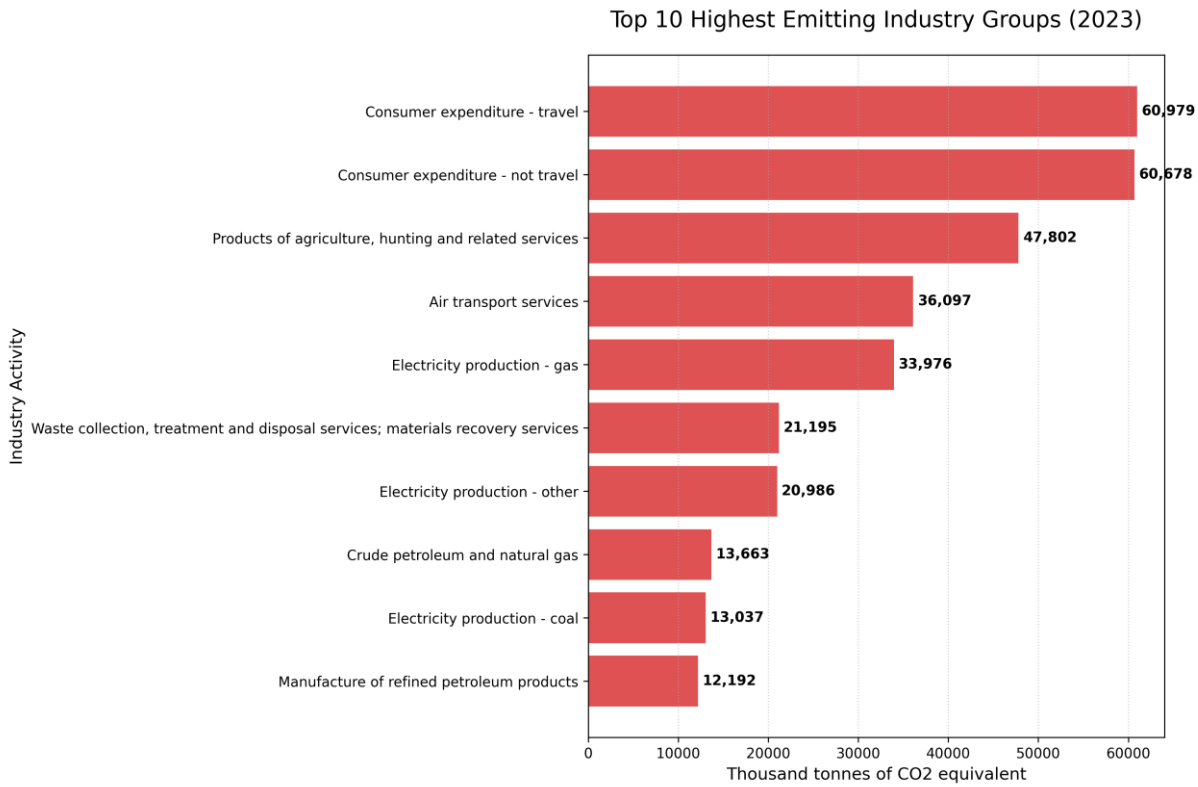


Fig 3: Top 10 highest-emitting industry groups

4. Implications for the UK Government

The UK has clearly been on the right path with steady reductions for the last three decades.

However, the projection shows that the **present rate of decrease is not sufficient to achieve net zero goals for 2050.**

Policy efforts should prioritise:

- decarbonising energy supply,
- cutting transport-related emissions,
- clean manufacturing, and
- influencing high-impact consumer activities.

5. Limitations and Improvements

Current Limitations

- We have used only historical emissions data in the model.

- We are not considering other major Factors like Technology advancements, policy changes, energy demand, and economic situations.
- The forecasting method is simple. Holt's model works well for showing long-term trends, but it cannot properly capture complex relationships between different sectors. For example, it does not show how an increase in manufacturing activity can generate higher emissions from freight transport. Because of this, some real-world scenarios are not fully reflected in the model.

Possible Improvements

- We can add other factors like **GDP, energy usage, and policy indicators**. Additionally, this would enable more **dynamic visualisations**, producing deeper insights that might enhance government decision-making.
- We can use more advanced models alongside Holt's method. Holt's model is good for simple trends, but it struggles with more complex patterns.
    - Machine learning approaches such as **Random Forest regression or Long Short-Term Memory (LSTM)** neural networks could be used to capture how different industries affect each other.
    - For example, they could show how growth in manufacturing leads to higher transport emissions. This would improve both the quality of the forecasts and the value of the analysis for policy planning.

Conclusion

UK greenhouse gas emissions have fallen significantly since 1990 and are forecast to continue declining. However, the projected pace of reduction suggests that **stronger and more targeted action is required** if the UK is to meet its net-zero target by 2050, particularly within the highest-emitting sectors identified in Figures 2 and 3.